ModelIt!

Student Activity Pack

How Does a Cut Heal Itself?

How Your Body Orchestrates Cell Division to Repair Damaged Tissue

Grade 08L2 | NGSS MS-LS1-4, MS-LS1-2 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What happens when you get a cut? Describe what you think is happening inside your body.

2 Why do some cuts heal faster than others?

3 What is cell division, and why might it be important for healing?

4 Draw a simple diagram showing what you think happens to the cells around a wound over time.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Wound Severity | Blood Supply | Cell Division Rate | New Tissue Formation | Healing Progress

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Minor vs. Deep

Low severity vs. high severity with normal blood supply

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Good vs. Poor Blood Supply

Same wound severity with high vs. low blood supply

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Worst Case

High severity with low blood supply

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Immune Response — White blood cells that fight infection at the wound site — infection diverts resources from repair

Scar Tissue Formation — Collagen-rich tissue that forms when wounds are too large for perfect regeneration

Growth Factor Signaling — Chemical signals released by damaged cells that trigger nearby cells to divide faster

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does your body know when you're injured, and how does it rebuild damaged tissue cell by cell? Use evidence from your simulation to support your answers.

1. **Why do wounds on the tongue or inside the mouth heal faster than wounds on the skin?**

2. **How does the body 'know' when to stop dividing cells once a wound is healed?**

3. **Why do older people generally heal more slowly than younger people?**

4. **How does understanding wound healing connect to understanding cancer (uncontrolled cell division)?**

5. **What would happen if cell division at a wound site never stopped?**

8 STEM Challenge: Design a Wound Treatment Protocol

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YOUR ENGINEERING MISSION

Design an evidence-based wound treatment plan that optimizes healing conditions, using your model data to justify why each treatment step accelerates tissue repair.

THE SCENARIO

A school nurse needs a science-based protocol for treating different types of student injuries — from minor scrapes to deeper cuts. Your team must create treatment guidelines that maximize healing speed based on the biological factors in your model.

Engineering Design Thinking

1. WHAT SPECIFIC CONDITIONS OPTIMIZE CELL DIVISION RATE AT THE WOUND SITE?
2. HOW WILL YOUR PROTOCOL ADDRESS BLOOD SUPPLY TO THE WOUND?
3. WHAT STEPS PREVENT INFECTION, WHICH WOULD DIVERT RESOURCES FROM HEALING?
4. HOW WILL YOU USE MODEL EVIDENCE TO EXPLAIN EACH TREATMENT RECOMMENDATION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Biomedical Researchers and Wound Care Specialists study how the body repairs itself and develop treatments that accelerate healing. They work in hospitals, research labs, and pharmaceutical

companies, earning $65,000-$120,000/year.